

Alex (Rui) Gan

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Education

University of Toronto - St. George

Honours Bachelor of Science

Computer Science Specialist, Focus in Artificial Intelligence

Sep 2018 – Sep 2022

Experience

Amazon

Toronto, ON

Software Development Engineer II

Aug 2022 – Present

- Led the design and implementation of the Books Ads **Agentic AI Platform** built on the Model Context Protocol (**MCP**) open standard and Bedrock, empowering teams to make data-driven decisions and automate complex tasks across distributed systems. Reduced advertiser insights analysis time from 2 days to minutes, increasing AE productivity by **20-30x** and driving **\$6MM annual revenue**.
- Designed and built Ad Pricing **Skill** on **AI Agent** with a secure cross-service data access across **4 auth systems** (Cognito, IAM, Redshift, Lake Formation). Reduced pricing effort from **2 hours to 3 minutes** per deal for business development teams.
- Built end-to-end data pipeline using Airflow and **Apache Spark** processing **35 billion** keywords daily across 7 marketplaces into DynamoDB. Improved book advertiser keyword coverage from 45% to **80%**, ROAS by **71%**, and projected **\$5MM ARR**.
- Led end-to-end development of **Amazon Posts and Post Boost**, coordinating **11+ teams** to enable one-click conversion of author content into Sponsored Brands ads. Reduced ad creation time by **70%**, with Boosted Post Ads increasing ROAS by 50%.
- Built automated oncall tool to backfill missing data and enable internal API calls, reducing customer issue resolution time by **10x** and reducing microservices integration effort by **60x**. Won organization's All Hands **Customer Obsession award**.

Amazon

Toronto, ON

Software Development Engineer Intern

May 2021 – Jul 2021

- Built operational excellence tool with **REST APIs** using **React** and **AWS** services (Lambda, API Gateway, SQS, CloudWatch, S3), enabling teams to build internal tools more efficiently.

Trinity College Chinese Students Association, University of Toronto

Toronto, ON

Web Developer

Jun 2020 – Apr 2021

- Developed club website for event information and member engagement to promote diversity.

Skills

Programming:	Java, Python, JavaScript/TypeScript, Scala, SQL, HTML/CSS, C, C#
Frameworks & Tools:	Apache Spark, Docker, React, Node.js, Express, Spring, Git, Linux, MongoDB
Cloud:	AWS (Lambda, S3, DynamoDB, CloudWatch, CDK, ECS, Redshift, Glue, Athena)
AI/ML:	Bedrock AgentCore, Claude, LLM Orchestration, Prompt/Spec Driven Development

Achievements and Awards

- Amazon Ads Org Customer Obsession Award
- School champion of the CEMC mathematics contests (Pascal and Gauss), scoring 142/150.